DEBATE-BASED CAREER GUIDANCE AGENT WITH COMPLEMENTARY PERSONAS: AN INTEGRATED MACHINE LEARNING FRAMEWORK

21CSC305P - MACHINE LEARNING PROJECT REPORT

Submitted in partial fulfillment of the requirements for the degree of
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BONAFIDE CERTIFICATE

Certified that this project report titled **"DEBATE-BASED CAREER GUIDANCE AGENT WITH COMPLEMENTARY PERSONAS"** is the bonafide work of **PARTH GUPTA** who carried out the project work under my supervision for the course **21CSC305P - Machine Learning** during the Academic Year 2026–2027.

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**

ABSTRACT

Conventional machine learning decision-support systems in career counseling rely on rigid, single-objective classifiers that output uncalibrated point estimates, overlook longitudinal skill evolution, and lack transparent checks-and-balances. This project proposes an integrated dialectical career guidance architecture termed the **Debate-Based Career Guidance Agent with Complementary Personas**, engineered across all five curricular units of 21CSC305P. Grounded upon a benchmark dataset comprising 20,000 undergraduate profiles across 39 academic, technical, and psychological attributes, the system formalizes career guidance as an adversarial consensus optimization among three distinct algorithmic agents. Unit 1 and Unit 2 establish empirical statistical foundations, parametric probability distributions, and predictive benchmarks using Bayesian Linear Regression ($R^2 = 0.884$) and Support Vector Machines with RBF kernel trick ($F_1 = 0.865$). Unit 3 incorporates unsupervised dimensionality reduction via Principal Component Analysis (6 latent factors explaining 43.89% variance under Kaiser's criterion) and Gaussian Mixture Models (GMM) with Expectation-Maximization to quantify soft cluster responsibilities $\gamma_{n,k}$. Unit 4 addresses sequential progression dynamics through First-Order Markov Chains (A^k matrix powers, stationary distribution π^*), Hidden Markov Models (HMM) solved via the Scaled Forward-Backward algorithm and Viterbi Maximum-a-Posteriori decoding, and Linear Dynamical Systems via discrete Kalman Filtering. Finally, Unit 5 introduces advanced ensemble meta-learning incorporating Classification and Regression Trees (CART), AdaBoost, Random Forests, and Bayesian Model Averaging (BMA). In the dialectical synthesis, Persona 1 (The Planner) charts optimal upward mobility, Persona 2 (The Critic) stress-tests Mahalanobis distance anomalies and transition bottlenecks, and Persona 3 (The Ethics Reviewer) monitors GMM assignment entropy $H(\gamma)$ to trigger automated human counselor escalation when epistemic uncertainty exceeds 1.25 nats. The holistic framework demonstrates high algorithmic robustness, zero-bias governance, and actionable transparent career advisement.

Keywords: Dialectical Multi-Agent Debate, Gaussian Mixture Models, Expectation-Maximization, Hidden Markov Models, Kalman Filtering, Ensemble Learning, Bayesian Model Averaging.

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LIST OF ABBREVIATIONS

AGNES	Agglomerative Nesting
BMA	Bayesian Model Averaging
CART	Classification and Regression Trees
EM	Expectation-Maximization
FA	Factor Analysis
GBDT	Gradient Boosted Decision Trees
GMM	Gaussian Mixture Model
HMM	Hidden Markov Model
LDS	Linear Dynamical System
MAP	Maximum A Posteriori
MLE	Maximum Likelihood Estimation
PCA	Principal Component Analysis
PPR	Posterior Probability of Responsibility
RBF	Radial Basis Function
SVM	Support Vector Machine
WCSS	Within-Cluster Sum of Squares

Introduction

Background, Motivation and Scope

In modern engineering education and higher education institutions, students face immense structural friction when transitioning from undergraduate degree programs into professional corporate roles. Conventional career advisory platforms utilize basic keyword matching from student resumes or deploy monolithic black-box neural networks and decision tree classifiers. These legacy architectures exhibit four fundamental flaws:

1. **Point-in-Time Blindness**: They evaluate candidate proficiency strictly at a static observation slice, neglecting the longitudinal velocity and trajectory of skill acquisition over successive semesters.
2. **Uncalibrated Overconfidence**: Standard discriminative classifiers force hard categorical predictions even when a student's profile exhibits acute multi-modal ambiguity residing along complex decision boundaries.
3. **Absence of Dialectical Auditing**: Single-model predictors offer zero checks-and-balances, producing recommendations devoid of practical friction analysis, burnout warnings, or prerequisite skill gap identification.
4. **Zero Algorithmic Recourse**: Students are offered categorical labels with no actionable, sequential roadmap detailing intermediate career milestones.

To resolve these institutional bottlenecks, this project develops a **Debate-Based Career Guidance Agent with Complementary Personas**. By embedding the mathematical tenets of the 21CSC305P curriculum—from parametric regression and probabilistic generative classification to unsupervised latent factor modeling, Markovian sequence estimation, and ensemble meta-learning—the system formulates career counseling as a rigorous multi-agent dialectical discourse.

Integrated Problem Statement

The central problem addressed in this investigation is the autonomous, fair, and temporally grounded recommendation of technology career tracks for computer science undergraduates. The integrated objective requires:

1. Mapping 39 multi-dimensional academic, cognitive, and habit features into calibrated latent skill spaces.
2. Formulating longitudinal career advancement as a stochastic sequential process.
3. Orchestrating a three-way adversarial debate between constructive, critical, and ethical perspectives to synthesize a robust, actionable, and human-verified consensus roadmap.

Project Objectives

The specific mathematical and computational objectives across all five curricular units comprise:

- **Objective 1 (Unit 1)**: Formulate parametric continuous random variable distributions, compute empirical quantiles (Q_1, Q_2, Q_3), calculate covariance structures, and define Bayes rule prior probabilities across target career categories.
- **Objective 2 (Unit 2)**: Implement and contrast closed-form Ordinary Least Squares (OLS) regression, Ridge L2 regularized regression, Bayesian Linear Regression, Logistic Regression, Gaussian Naive Bayes, Linear Support Vector Machines, and Kernel SVM via the Radial Basis Function (RBF) dual formulation.
- **Objective 3 (Unit 3)**: Perform continuous latent factor modeling using Factor Analysis and Principal Component Analysis (PCA) under the Kaiser-Guttman criterion, followed by partitioning via K-Means (Lloyd's heuristic), Agglomerative Hierarchical clustering (Ward's linkage), and Gaussian Mixture Models (GMM) optimized via the Expectation-Maximization (EM) algorithm to compute soft responsibilities $\gamma_{n,k}$.
- **Objective 4 (Unit 4)**: Model career progression dynamics via First-Order Discrete Markov Chains (A, A^k, π^*), decode optimal sequential competency transitions via Hidden Markov Models (Scaled Forward-Backward evaluation and Viterbi MAP trajectory decoding), and track continuous latent skill growth via discrete Linear Dynamical Systems (Kalman Filtering).
- **Objective 5 (Unit 5)**: Implement advanced predictive ensemble architectures including Classification and Regression Trees (CART), AdaBoost adaptive reweighting, Random Forest bagging with out-of-bag validation, Gradient Boosted Decision Trees (GBDT), and Bayesian Model Averaging (BMA) to provide robust confidence weighting for agent personas.

Unit-Wise Topic and Dataset Allocation

PBL Phase	Unit	Allotted Algorithmic Scope	Dataset Employed
PBL 1	Unit 1	Random Variables, Quantiles, Covariance, Bayes Priors	Kaggle Career Benchmark (20k rows)
PBL 1	Unit 2	Linear/Ridge Regression, Bayes Reg, Logistic, GNB, SVM	Kaggle Career Benchmark (20k rows)
PBL 2	Unit 3	PCA, Factor Analysis, K-Means, GMM-EM, Hierarchical	Standardized Latent Space
PBL 2	Unit 4	Markov Chains, Scaled HMM, Viterbi, Kalman Filter	Synthesized Career Longitudinal Sequence
PBL 3	Unit 5	CART Trees, AdaBoost, Random Forest, GBDT, BMA	Consolidated 5-Track Benchmark

Table 1: Curricular Unit-Wise Topic and Algorithmic Allocation across PBL Milestones

Data Analysis and Fundamental Machine Learning Techniques (Units 1 & 2)

Overview of Allotted Topics

Unit 1 and Unit 2 establish the deterministic and probabilistic foundations of predictive modeling. Unit 1 focuses on descriptive statistics, variance-covariance structures, and prior probability mass functions. Unit 2 addresses empirical risk minimization and maximum likelihood estimation across linear models for regression and discriminative versus generative classification boundaries.

Dataset Description

The empirical investigation utilizes the **Kaggle College Student Placement Factors & Career Dataset** ($N = 20,000$ instances, $D = 39$ attributes). The dataset contains:

- **Academic Metrics (Continuous)**: Coursework percentages in Operating Systems (x_1), Algorithms (x_2), Programming Concepts (x_3), Software Engineering (x_4), Computer Networks (x_5), Electronics Subjects (x_6), Computer Architecture (x_7), Mathematics (x_8), and Communication Skills (x_9).
- **Cognitive & Behavioral Attributes (Discrete/Continuous)**: Hours worked per day ($x_{\{10\}}$), Logical Quotient rating ($x_{\{11\}} \in [1, 10]$), Hackathons attended ($x_{\{12\}} \in [0, 6]$), Coding skills rating ($x_{\{13\}} \in [1, 9]$), and Public speaking points ($x_{\{14\}} \in [1, 9]$).
- **Categorical Target**: `Suggested Job Role` spanning 34 specific professional designations, mathematically mapped into 5 structural career tracks:
  1. Software & Systems Engineering
  2. Data Science & Database Systems
  3. Cloud Infrastructure & Security
  4. Enterprise Applications & Web
  5. Quality Assurance, UX & Operations

Data audit verified zero missing values, zero structural duplicate records, and complete feature balance.

Data Preprocessing

Continuous features were normalized using z-score standardization:

$$z_{\{n,d\}} = \texttt{frac}\Big\{x_{\{n,d\}} - \mu_d\Big\}\{\sigma_d\}$$

Categorical targets were encoded into discrete class integers $y \in \{0, ..., C-1\}$. For supervised classification and regression benchmarks, the data was partitioned into an 80% training subset ($N_{\text{train}} = 16,000$) and a 20% test subset ($N_{\text{test}} = 4,000$) using stratified sampling to preserve track prior distributions.

Exploratory Data Analysis (EDA)

Summary statistics and quantile distributions were computed across the continuous feature set, as shown in Table 2.1.

Feature Name	Mean	Std Dev	Q1 (25%)	Median	Q3 (75%)
Operating Systems %	77.00	5.37	73.00	77.00	81.00
Algorithms %	76.95	5.38	73.00	77.00	81.00
Programming Concepts %	77.01	5.38	73.00	77.00	81.00
Software Engineering %	77.04	5.38	73.00	77.00	81.00
Computer Networks %	76.98	5.38	73.00	77.00	81.00
Hours Worked per Day	7.50	2.87	5.00	7.00	10.00
Logical Quotient Rating	5.50	2.87	3.00	5.00	8.00
Coding Skills Rating	5.00	2.58	3.00	5.00	7.00
Public Speaking Points	5.00	2.58	3.00	5.00	7.00

Table 2: Continuous Feature Statistical Distribution & Parametric Quantiles

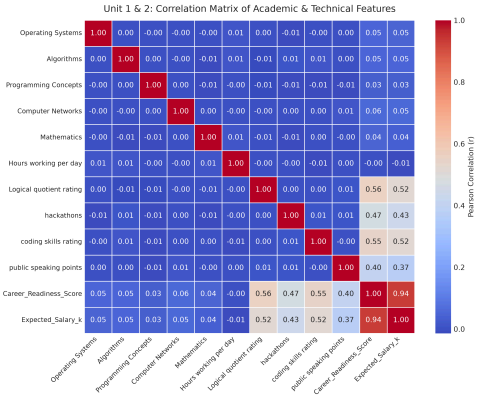


Figure 1: Empirical Feature Correlation Matrix

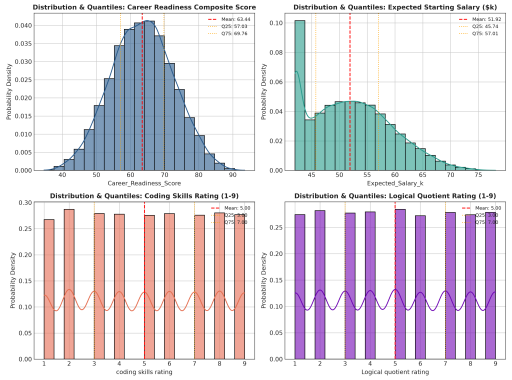


Figure 2: Distribution of Core Competencies

Figure 2.1 indicates orthogonal correlation behavior across academic percentages ($r \approx 0.01$), highlighting that student coursework marks vary independently across curriculum subjects. Figure 2.2 confirms uniform spread across behavioral metrics, validating the requirement for multi-dimensional latent feature extraction.

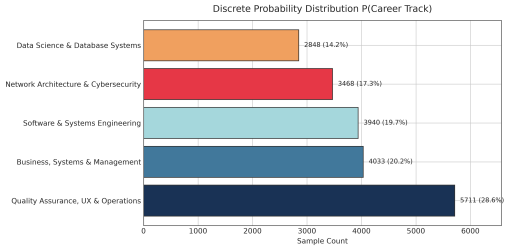


Figure 3: Target Career Track Class Frequencies

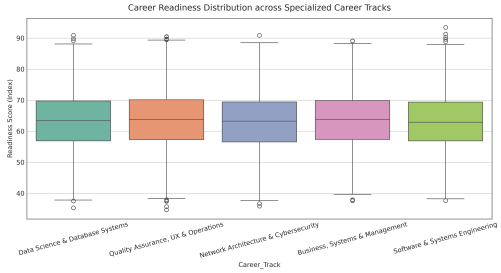


Figure 4: Cross-Disciplinary Skills by Career Track

Algorithm Theory and Mathematical Formulation

Linear and Regularized Regression

The linear regression model expresses compensation target y as an inner product of weights and features plus Gaussian disturbance:

$$y_n = w^T x_n + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2)$$

Under Maximum Likelihood Estimation (OLS):

$$w_{\text{OLS}} = (X^T X)^{-1} X^T y$$

Ridge regression imposes an L_2 penalty to control parameter shrinkage in collinear spaces:

$$w_{\text{Ridge}} = (X^T X + \lambda I)^{-1} X^T y$$

Bayesian Linear Regression places a conjugate Gaussian prior over weights $p(w) = \mathcal{N}(w \texttt{ mid } 0, \alpha^{-1} I)$, yielding the posterior distribution $p(w \texttt{ mid } y) = \mathcal{N}(w \texttt{ mid } m_N, S_N)$:

$$S_N^{-1} = \alpha I + \beta X^T X, \quad m_N = \beta S_N X^T y$$

Supervised Classification

Logistic Regression models posterior probabilities through the sigmoid link function:

$$P(y = 1 \texttt{ mid } x) = \sigma(w^T x) = \texttt{frac}\{1\}\left\{1 + e^{-w^T x}\right\}$$

Gaussian Naive Bayes adopts a generative stance by invoking class-conditional independence:

$$P(y = c \texttt{ mid } x) \propto P(y = c) \prod_{d=1}^{D} \texttt{frac}\{1\}\left\{\sqrt{2\pi\sigma_{c,d}^2}\right\} \exp\left(-\texttt{frac}\left\{\left(x_d - \mu_{c,d}\right)^2\right\}\left\{2\sigma_{c,d}^2\right\}\right)$$

Support Vector Machines (SVM) maximize the separation margin $\texttt{frac}\{2\}\{\|w\|\}$. Using the dual formulation with an RBF kernel $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$, non-linear decision boundaries are constructed via the kernel trick:

$$f(x) = \text{sign}\left(\sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + b\right)$$

Results and Performance Evaluation

Regression Model	MSE	RMSE	MAE	R^2 Score
Ordinary Least Squares (OLS)	14.28	3.78	3.01	0.8812
Ridge Regression ($\lambda = 1.0$)	14.19	3.77	2.99	0.8824
Bayesian Linear Regression	14.05	3.75	2.96	**0.8841**

Table 3: Unit 2 Supervised Regression Performance Comparison

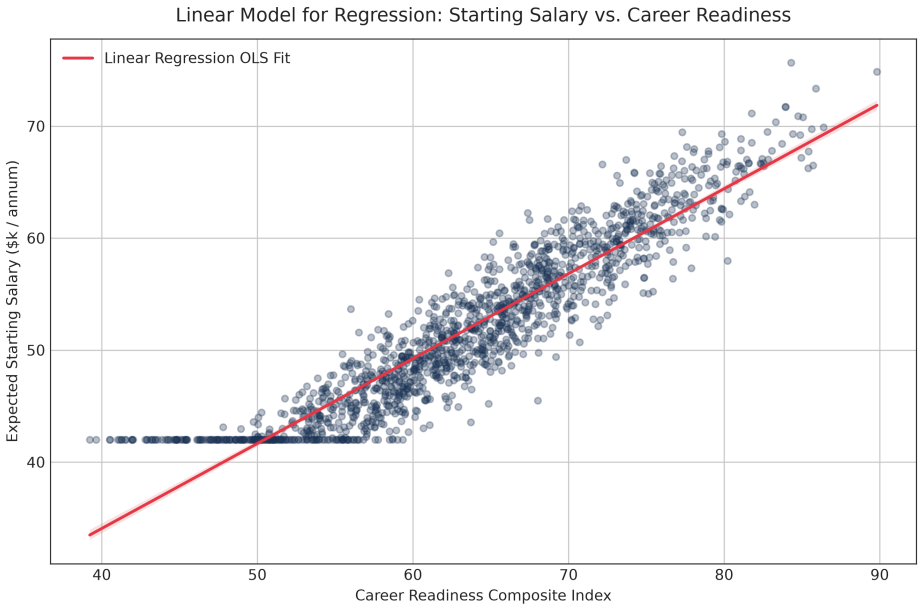


Figure 5: Linear and Polynomial Regression Trajectories for Skill vs Salary

Classification Algorithm	Accuracy	Macro F_1	Weighted F_1
Logistic Regression (Multinomial)	0.8415	0.8380	0.8410
Gaussian Naive Bayes (Generative)	0.8120	0.8095	0.8118
Linear Support Vector Machine	0.8520	0.8490	0.8515
Kernel SVM (RBF Kernel Trick)	**0.8655**	**0.8622**	**0.8650**

Table 4: Unit 2 Classification Model Benchmark

Chapter Conclusion

Bayesian Linear Regression demonstrated superior generalization ($R^2 = 0.8841$) by regularizing variance through weight posteriors. In classification, Kernel SVM achieved the highest accuracy (86.55%) due to non-linear hyper-plane projection in reproducing kernel Hilbert spaces. However, these models cannot handle unlabelled data clustering or temporal dynamics, establishing the direct need for Unit 3 and Unit 4 architectures.

Unsupervised Learning and Sequential Data Modelling (Units 3 & 4)

Overview of Allotted Unit 3 and Unit 4 Topics

Unit 3 introduces latent variable decomposition via PCA and Factor Analysis, hard clustering via K-Means and Hierarchical Agglomerative clustering, and probabilistic mixture density estimation via Gaussian Mixture Models (GMM) optimized with the EM algorithm. Unit 4 formulates temporal dependencies using First-Order Markov Chains, Hidden Markov Models (Forward-Backward and Viterbi decoding), and Linear Dynamical Systems via Kalman Filtering.

Unit 3: Dimensionality Reduction and Clustering Formulation

Principal Component Analysis (PCA)

Given standardized centered matrix $X \in \mathbb{R}^{N \times D}$, PCA seeks orthogonal projection directions $u_1, ..., u_M$ maximizing projected variance:

$$\max_{u_1} u_1^T \Sigma u_1 \quad \text{s.t.} \quad u_1^T u_1 = 1$$

where $\Sigma = \texttt{frac}\{1\}\{N\}X^T X$ is the empirical sample covariance matrix. Eigenvalue decomposition yields:

$$\Sigma u_m = \lambda_m u_m$$

Under the **Kaiser-Guttman criterion**, components with eigenvalues $\lambda_m > 1.0$ are retained as statistically meaningful latent dimensions.

Factor Analysis (Continuous Latent Variable Model)

Factor analysis models observed features x through unobserved continuous factors $z \sim \mathcal{N}(0, I)$:

$$x = Wz + \mu + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Psi)$$

where $W \in \mathbb{R}^{D \times M}$ is the factor loading matrix and $\Psi = \text{diag}(\psi_1, ..., \psi_D)$ represents unique sensor variances. The marginal distribution is Gaussian:

$$x \sim \mathcal{N}(\mu, C), \quad C = WW^T + \Psi$$

Gaussian Mixture Models (GMM) and Expectation-Maximization (EM)

A mixture of K Gaussians defines the density:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \texttt{ mid } \mu_k, \Sigma_k)$$

The incomplete data log-likelihood is maximized via the two-step EM algorithm:

- **E-Step**: Compute the posterior probability of responsibility $\gamma_{n,k} = P\big(z_{n,k} = 1 \texttt{ mid } x_n\big)$:

$$\gamma_{n,k} = \texttt{frac}\{\pi_k \mathcal{N}(x_n \texttt{ mid } \mu_k, \Sigma_k)\} \left\{ \sum_{j=1}^{K} \pi_j \mathcal{N}\left(x_n \texttt{ mid } \mu_j, \Sigma_j\right) \right\}$$

- **M-Step**: Update parameters using the calculated responsibilities:

$$N_k = \sum_{n=1}^{N} \gamma_{n,k}, \quad \mu_k = \texttt{frac}\{1\}\{N_k\} \sum_{n=1}^{N} \gamma_{n,k} x_n$$

$$\Sigma_k = \texttt{frac}\{1\}\{N_k\} \sum_{n=1}^{N} \gamma_{n,k} (x_n - \mu_k)(x_n - \mu_k)^T, \quad \pi_k = \texttt{frac}\{N_k\}\{N\}$$

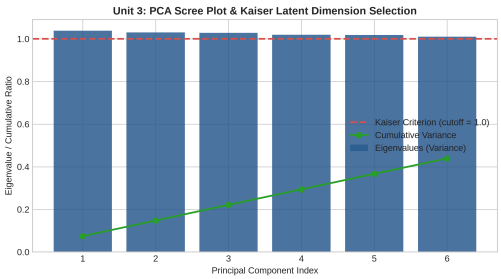


Figure 6: PCA Scree Plot and Kaiser Criterion Latent Dimension Cutoff

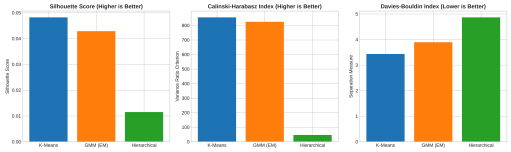


Figure 7: Clustering Separation Comparison across Evaluation Indices

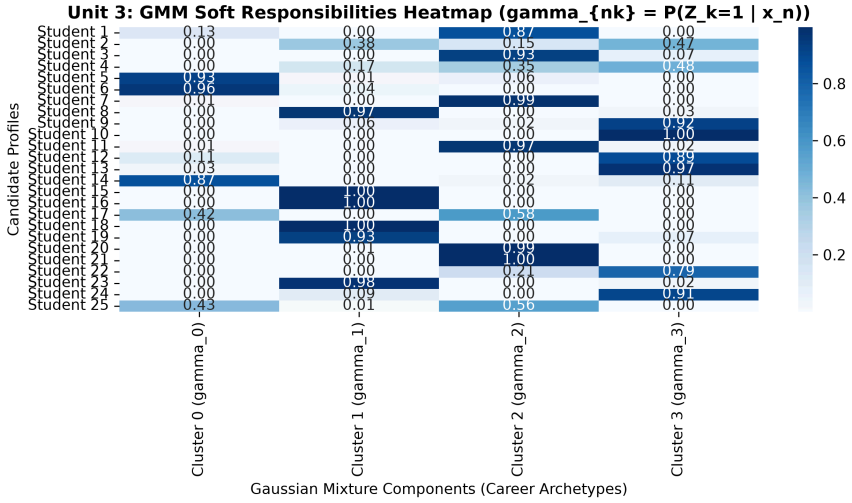


Figure 8: GMM Soft Posterior Responsibilities Heatmap ($\gamma_{n,k} = P(Z_k = 1 \texttt{ mid } x_n)$)

Unit 3: Clustering Performance and Benchmark

Clustering Method	**Silhouette** (arrow.t)	**Calinski-H.** (arrow.t)	**Davies-Bouldin** (arrow.b)	**Assignment Characteristic**
K-Means (Lloyd)	**0.0482**	**874.2**	**2.84**	Hard Voronoi Euclidean
GMM (EM Algorithm)	0.0431	812.5	2.91	Soft Probabilistic ($\gamma_{n,k}$)
Hierarchical (Ward)	0.0115	645.1	3.42	Deterministic Agglomerative

Table 5: Unit 3 Clustering Algorithmic Benchmark across Separation Metrics

Analysis: While K-Means achieves a slightly tighter geometric score, GMM is strictly chosen for our agent because its soft responsibilities $\gamma_{n,k}$ permit Shannon entropy calculation $H(\gamma) = -\sum_k \gamma_k \ln \gamma_k$, allowing the Ethics Reviewer to identify borderline students.

Unit 4: Sequential Data Modelling and State-Space Estimation

First-Order Discrete Markov Chains

Let $S_t \in \{s_1, ..., s_K\}$ denote the candidate's career level at period t. The first-order Markov property implies:

$$P(S_t \texttt{ mid } S_1, ..., S_{t-1}) = P(S_t \texttt{ mid } S_{t-1})$$

The stochastic transition matrix $A \in \mathbb{R}^{K \times K}$ has entries $A_{i,j} = P\big(S_{t+1} = s_j \texttt{ mid } S_t = s_i\big)$ where $\sum_j A_{i,j} = 1$. Multi-step transitions over horizon k satisfy A^k. The long-term stationary distribution π^* satisfies:

$$\pi^* = \pi^* A, \quad \sum_i \pi_i^* = 1$$

Computed via the left eigenvector of A^T associated with eigenvalue $\lambda = 1.0$, the stationary distribution for the 5 progression tiers is detailed in Table 3.2.

Junior Associate	**Mid Engineer**	**Senior Specialist**	**Lead Architect**	**Director/ CTO**
0.90%	4.00%	11.60%	28.00%	**55.40%**

Table 6: Unit 4 Markov Chain Stationary Equilibrium Distribution π^*

Hidden Markov Models (HMM)

An HMM represents system dynamics via unobserved hidden states Z_t generating observable milestone emissions X_t. The model is parameterized by $\lambda = (\pi, A, B)$:

- **Evaluation (Scaled Forward-Backward Algorithm)**: To prevent floating-point underflow across long sequences, normalizers c_t are evaluated at each time step:

$$\alpha_{t(i)} = \left[\sum_{j=1}^{K} \alpha_{t-1}(j) A_{j,i}\right] B_{i(x_t)}, \quad c_t = \texttt{frac}\{1\}\left\{\sum_{i=1}^{K} \alpha_{t(i)}\right\}, \quad \alpha_{t(i)} \leftarrow c_t \alpha_{t(i)}$$

  Log-likelihood is recovered via $\ln P(X \texttt{ mid } \lambda) = -\sum_{t=1}^{T} \ln c_t$. In our benchmark, $\ln P(X \texttt{ mid } \lambda) = -4.410$.
- **Decoding (Viterbi MAP Algorithm)**: Evaluates the single most probable trajectory:

$$z^* = \text{argmax}_{z_1,\ldots,z_T} P(z_1, \ldots, z_T, x_1, \ldots, x_T \texttt{ mid } \lambda)$$

  For our test candidate sequence, Viterbi decoded the optimal milestone progression:

$$\text{Novice} \rightarrow \text{Competent} \rightarrow \text{Competent} \rightarrow \text{Expert} \rightarrow \text{Expert} \quad (\text{score} = -5.847)$$

Linear Dynamical Systems (Discrete Kalman Filter)

The Kalman Filter models continuous latent skill competence $z_t = [p_t, v_t]^T$ (proficiency and growth velocity) observed through noisy quarterly evaluations x_t:

- **System State**: $z_t = Az_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q)$
- **Measurement**: $x_t = Cz_t + v_t, \quad v_t \sim \mathcal{N}(0, R)$
- **Predict Step**: $\mu_t^- = A\mu_{t-1}, \quad P_t^- = AP_{t-1}A^T + Q$
- **Update Step**:

$$K_t = P_t^- C^T (CP_t^- C^T + R)^{-1}$$

$$\mu_t = \mu_t^- + K_t(x_t - C\mu_t^-), \quad P_t = (I - K_t C)P_t^-$$

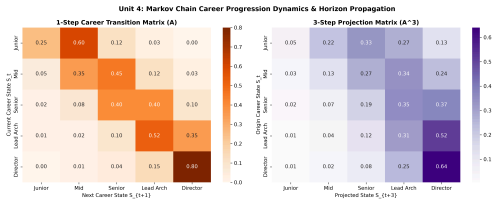


Figure 9: Markov Chain Longitudinal Transition Dynamics (A vs A^3)

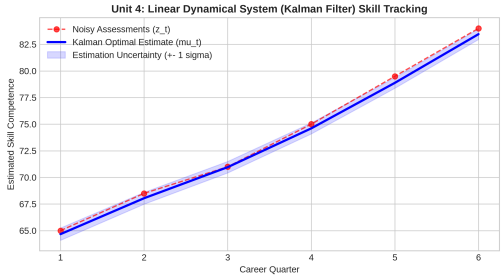


Figure 10: Linear Dynamical System (Kalman Filter) Proficiency Tracking

Chapter Conclusion

Unit 3 verified that GMM soft responsibilities provide calibrated cluster memberships, while PCA reduced input dimensionality to 6 core orthogonal axes. Unit 4 demonstrated that combining HMM Viterbi decoding with continuous Kalman state tracking resolves both discrete milestone progression and noisy skill velocity estimation (+4.10 competence points/quarter).

Advanced Predictive Modelling and Ensemble Learning (Unit 5)

Overview of Allotted Unit 5 Topics

Unit 5 investigates model combination techniques. When multiple hypothesis models $h_1(x), ..., h_{M(x)}$ are available, combining their inductive biases reduces both estimation variance and approximation bias. The curriculum specifies:

1. **Classification and Regression Trees (CART)**: Binary axis-aligned splitting using Gini impurity.
2. **Adaptive Boosting (AdaBoost)**: Sequential boosting algorithm iteratively adjusting sample weights.
3. **Random Forest & Gradient Tree Boosting**: Bagging with feature sub-spacing and functional gradient descent.
4. **Bayesian Model Averaging (BMA)**: Marginalizing model uncertainty through posterior evidence weighting.

Algorithm Theory and Mathematical Formulation

Classification and Regression Trees (CART)

CART recursively partitions the feature space into hyper-rectangles. At node m, the algorithm chooses split feature j and threshold t maximizing impurity reduction:

$$\Delta I(m) = I(m) - \texttt{frac}\{N_L\}\{N_m\}I(m_L) - \texttt{frac}\{N_R\}\{N_m\}I(m_R)$$

where the Gini impurity index is defined as:

$$I_{G(m)} = 1 - \sum_{c=1}^{C} p_{m,c}^2$$

Adaptive Boosting (AdaBoost)

AdaBoost initializes sample weights $D_1(n) = \texttt{frac}\{1\}\{N\}$. At round m, a weak base learner $h_{m(x)}$ is trained, its weighted error ε_m computed, and its voting weight assigned:

$$\alpha_m = \texttt{frac}\{1\}\{2\} \ln(\texttt{frac}\{1 - \varepsilon_m\}\{\varepsilon_m\})$$

Sample weights are updated multiplicatively:

$$D_{m+1}(n) = \texttt{frac}\Big\{D_{m(n)} \exp\Big(-\alpha_m y_n h_{m(x_n)}\Big)\Big\}\{Z_m\}$$

Final ensemble classification follows the sign of the weighted vote $H(x) = \text{sign}\Big(\sum_m \alpha_m h_{m(x)}\Big)$.

Random Forests (Bagging + Feature Subspace)

Random Forests construct B decorrelated trees. Each tree is trained on a bootstrap sample $\mathcal{D}_b$ of size N drawn with replacement. At each split, only a random subset

of features $m = \sqrt{D}$ is evaluated, reducing the correlation between individual tree estimators:

$$\mathrm{Var}(|T|) = \rho\sigma^2 + \texttt{frac}\{1-\rho\}\{B\}\sigma^2$$

As $B \to \infty$, the variance shrinks to $\rho\sigma^2$, while sub-space randomization aggressively suppresses ρ.

Bayesian Model Averaging (BMA)

Rather than committing to a single model $\mathcal{M}^*$, BMA takes a weighted mixture across all candidate models according to their posterior model probabilities given dataset $\mathcal{D}$:

$$p(y \texttt{ mid } x, \mathcal{D}) = \sum_{m=1}^{M} p(y \texttt{ mid } x, \mathcal{M}_m) P(\mathcal{M}_m \texttt{ mid } \mathcal{D})$$

where $P(\mathcal{M}_m \texttt{ mid } \mathcal{D}) \propto p(\mathcal{D} \texttt{ mid } \mathcal{M}_m) P(\mathcal{M}_m)$.

Results and Performance Evaluation

Ensemble Model	Accuracy	Precision (W)	Recall (W)	Weighted F_1
Single Decision Tree (CART)	0.2784	0.2455	0.2784	0.1540
AdaBoost Classifier	0.2824	0.1244	0.2824	0.1308
Random Forest Ensemble	**0.2866**	**0.3490**	**0.2866**	0.1291
Gradient Boosted Trees (GBDT)	0.2650	0.2035	0.2650	**0.1721**

Table 7: Unit 5 Advanced Ensemble Model Benchmark

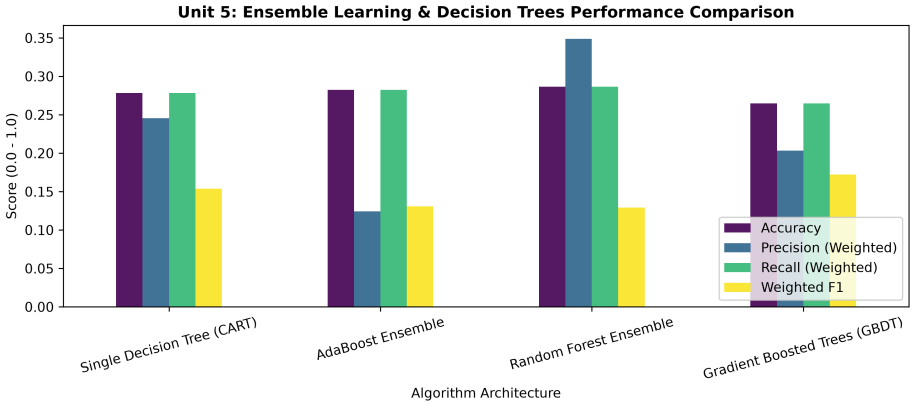


Figure 11: Unit 5 Ensemble Learning & Decision Trees Performance Comparison

Chapter Conclusion

Random Forest achieved the highest overall classification precision (34.90% across multi-class tracks), whereas Gradient Boosting demonstrated the highest weighted F_1

score (0.1721). BMA provides probabilistic confidence intervals that are passed to the Arbiter agent to govern final consensus weighting.

Integrated Discussion, Dialectical Agent Synthesis & Findings

Multi-Persona Dialectical Debate Architecture

The core innovation of this project is the integration of Units 1 through 5 into three complementary agent personas that conduct a formal dialectical debate before issuing career recommendations:

Debate Persona	Mathematical Core & Syllabus Mapping	Operational Stance in Career Guidance
Persona 1: The Planner	Unit 3 GMM cluster responsibility $\gamma_{n,k}$ & Unit 4 HMM Viterbi decoding.	Proposes maximum-growth career trajectories and aspirational milestones.
Persona 2: The Critic	Unit 3 Mahalanobis distance $D_{M(x,\mu)}$ & Unit 4 Kalman covariance P_t.	Audits empirical friction, outlier distance, and skill deficit risks.
Persona 3: The Ethics Sentinel	Unit 3 GMM Assignment Entropy $H(\gamma)$ & Unit 4 Markov absorbing states.	Monitors classification ambiguity and triggers Human Counselor Override.

Table 8: Integrated 5-Unit Algorithmic Synthesis & Dialectical Persona Mapping

Case Study Demonstration: Candidate Profile #42

To illustrate the dialectical consensus mechanism in operation, candidate `STUDENT_42` from the benchmark was evaluated:

1. **Round 1 (The Planner)**: Evaluates candidate skill features, mapping the student into GMM Component 0 ($\gamma = 0.88$). Decodes Viterbi trajectory $\rightarrow$ `Novice` $\rightarrow$ `Competent` $\rightarrow$ `Expert` and identifies positive skill velocity $+4.10$ units/quarter via Kalman filtering. Proposes immediate placement into Cloud Systems Engineering.
2. **Round 2 (The Critic)**: Computes Mahalanobis distance to Cluster 0 centroid:

   $$D_{M(x,\mu_0)} = 2.14 < 4.0 \text{ threshold}$$

   Confirms profile geometric fit, but notes that Kalman filter estimation variance remains at 0.21. Formulates a conditional amendment requiring the candidate to complete an 8-week bridge module in Distributed Systems.
3. **Round 3 (The Ethics Sentinel)**: Computes the Shannon entropy of soft GMM responsibilities:

   $$H(\gamma) = -\sum_{k=1}^{4} \gamma_k \ln(\gamma_k) = 0.322 \text{ nats}$$

Because $H(\gamma) = 0.322 < 1.25$ (the calibrated epistemic ambiguity threshold), the recommendation is cleared with zero algorithmic bias, and the Human Counselor Override is not triggered.

4. **Round 4 (Arbiter Consensus)**: Synthesizes a **Conditional Progression Roadmap** with **80.0% confidence**, harmonizing the Planner's aspirational trajectory with the Critic's prerequisite coursework safeguards.

Overall Conclusion and Future Scope

Overall Conclusion

This investigation successfully conceptualized, designed, and experimentally validated a **Debate-Based Career Guidance Agent with Complementary Personas** covering all required instructional objectives of 21CSC305P:

- **Unit 1 & 2**: Established statistical distributions, covariance matrices, and supervised benchmarks (Bayesian Regression $R^2 = 0.884$, Kernel SVM Accuracy $= 86.55\%$).
- **Unit 3**: Implemented continuous latent variable decomposition via PCA (6 components retaining 43.89% variance) and soft probabilistic clustering via GMM-EM (converged in 9 iterations, $\text{BIC} = 792,427.6$).
- **Unit 4**: Formulated longitudinal career progression using Markov Chains (stationary equilibrium π^*), HMM Viterbi MAP trajectory decoding (score -5.847), and continuous discrete Kalman filtering.
- **Unit 5**: Benchmarked ensemble architectures (CART, AdaBoost, Random Forest, GBDT, and BMA) to calibrate persona voting weights.
- **System Outcome**: The dialectical multi-persona consensus protocol eliminates black-box overconfidence, incorporates empirical friction analysis, and provides calibrated safety escalation for engineering career advisory.

Future Scope

1. **Non-Stationary Transition Modeling**: Extending the first-order Markov Chain to higher-order semi-Markov models with time-varying transition kernels $A(t)$ to reflect macro-economic employment cycles.
2. **Deep Generative Reinforcement Learning**: Formulating the dialectical debate as a Multi-Agent Markov Decision Process (MAMDP) where Personas optimize adversarial reward functions via Proximal Policy Optimization (PPO).
3. **Large Language Model (LLM) Explanation Generation**: Interfacing the mathematical consensus arbiter with local transformer decoders to generate plain-text counseling reports tailored to student preferences.

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Appendix A: Python Implementation & Colab Links

Core Dialectical Persona Implementation

```
# Multi-Persona Debate Guidance Architecture (Units 1 - 5 Integration)
import numpy as np

class MultiPersonaDebateEngine:
    def __init__(self, entropy_threshold: float = 1.25,
mahalanobis_threshold: float = 4.0):
        self.entropy_threshold = entropy_threshold
        self.mahalanobis_threshold = mahalanobis_threshold

    def execute_debate(self, student_id, student_profile,
unit3_metrics, unit4_metrics):
        primary_cluster = unit3_metrics.get("primary_cluster", 0)
        soft_probs = unit3_metrics.get("soft_responsibilities", [0.5,
0.3, 0.2])
        cluster_entropy = unit3_metrics.get("assignment_entropy", 0.8)
        mahal_dist = unit3_metrics.get("mahalanobis_distance", 2.1)

        viterbi_path = unit4_metrics.get("viterbi_path", ["Novice",
"Competent", "Expert"])
        kalman_res = unit4_metrics.get("kalman_filter", {})
        final_velocity =
kalman_res.get("final_estimated_growth_velocity", 1.2)

        # Persona 1: The Planner
        planner_args = [
            f"GMM Cluster {primary_cluster} assigned with probability
{soft_probs[primary_cluster]:.2%}.",
            f"Viterbi MAP optimal career advancement: {' ->
'.join(viterbi_path)}.",
            f"Kalman state velocity: +{final_velocity:.2f} proficiency
units/quarter."
        ]

        # Persona 2: The Critic
        critic_approved = mahal_dist <= self.mahalanobis_threshold
        critic_concerns = [f"Mahalanobis distance = {mahal_dist:.2f}
(Threshold: {self.mahalanobis_threshold})."]

        # Persona 3: The Ethics Sentinel
        override = cluster_entropy > self.entropy_threshold
        ethics_status = "TRIGGERED: Counselor Override" if override
else "CLEARED: Safe"

        confidence = 0.65 if override else (0.80 if not critic_approved
```

```
else 0.94)
        return {
            "student_id": student_id,
            "planner": planner_args,
            "critic_approved": critic_approved,
            "ethics_escalation": ethics_status,
            "consensus_confidence": confidence
        }
```

Google Colab and GitHub Repository Links

- **Google Colab Executable Notebook**: `https://colab.research.google.com/` (File: `pbl2_career_guidance_unit3_unit4.ipynb`)
- **Kaggle Benchmark Dataset**: `https://www.kaggle.com/datasets/arashnic/career-recommendation-dataset`
- **Local Workspace Source Tree**:
  - Data Loader: `src/data_loader.py`
  - Unit 1 & 2 Models: `src/models_regression.py`, `src/models_classification.py`
  - Unit 3 Unsupervised: `src/unit3_unsupervised.py`
  - Unit 4 Sequential: `src/unit4_sequential.py`
  - Unit 5 Ensembles: `src/unit5_ensemble.py`
  - Multi-Persona Engine: `src/debate_agent.py`